Knowing When You Do Not Know: Sequential Map-Anchored Visual Localization for GNSS-Denied UAV Flight

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***Abstract*—Matching a downward-looking camera against a pre-loaded satellite map is the standard route to absolute position when GNSS is denied. The recent literature optimises this as a single-image retrieval problem, and the common benchmark is close to saturated. A flying aircraft, however, produces a *sequence*, and in that setting the binding constraint is not retrieval accuracy but the fact that a large fraction of frames cannot be matched at all. We fuse cross-view matching with visual odometry in a particle filter and evaluate on ten real UAV survey flights spanning 406–2572 m altitude, 9–103 km per flight and acquisition dates from 2016 to 2023. On the seven flights whose imagery is usable against the basemap the system holds a median error of 8.4–24.8 m with $\geq$99.7% of frames localised, while visual odometry alone drifts 2803 m over 74 km. Two by-products follow from the same machinery: the residual rotation of each map match exposes the platform's heading error, letting the system calibrate its own magnetometer deviation without GNSS and cutting odometry drift from 3.795% to 0.865% of distance travelled; and the *match rate*, measurable on a planned route before flight, predicts whether the system will work there at all ($\rho = -0.764$ against log error, with a threshold near 50%). Finally we report a negative result that we believe generalises: replacing LoFTR with RoMa raises match rate on the failing flights from 12–42% to 96–100% and makes end-to-end accuracy dramatically *worse*, because RoMa returns confident correspondences for unrelated imagery. For localization, a matcher is useful in proportion to its ability to stay silent, not to match.**

***Index Terms*—GNSS-denied navigation, cross-view geo-localization, particle filter, visual odometry, unmanned aerial vehicles, image matching**

I. Introduction

An unmanned aerial vehicle whose GNSS receiver is jammed or spoofed loses absolute position. Inertial and visual odometry continue to supply relative motion, but their error accumulates without bound: on the 74 km flight studied here, frame-to-frame visual odometry ends 2803 m from truth. The established remedy is to carry a georeferenced satellite basemap and recover absolute position by matching what the camera sees against it.

Recent work treats this as cross-view image retrieval and has driven the common benchmark close to saturation; University-1652 now sits at 97.5% Recall@1 [1]. That framing quietly assumes the hard part is choosing correctly among candidate locations. In flight over real terrain we find a different constraint dominates: on a substantial fraction of frames *no* correct match exists to be chosen. Over water, uniform farmland, or repeated building patterns, and wherever the basemap and the camera disagree in season or in years, matching returns nothing usable. In our single-frame baseline this is 23% of frames on the easiest flight in our set and 96% on the hardest.

A flying aircraft is not a collection of independent photographs. It follows a trajectory, and frames that cannot be matched are bracketed by frames that can. We therefore treat the problem as sequential state estimation: visual odometry supplies a motion model, map matching supplies absolute fixes and is permitted to fail, and a particle filter carries the two together. This is standard practice in ground robotics but is almost absent from the recent UAV cross-view literature, which we survey in Section II.

Our contributions are:

1) A sequential map-anchored localization system evaluated on *ten* real UAV flights rather than one, spanning a 6.3× altitude range and seven years of acquisition dates, fully automatic with per-flight calibration derived from each flight's own first 20% and evaluated on the remainder (Section V).
2) Online self-calibration of platform heading and image scale from the geometry of the map matches themselves, requiring no GNSS. The measured magnetometer deviation, $-1.93°$ and $-7.08°$ on the two legs of a survey pattern, agrees with the GNSS-derived value in magnitude and sign, and correcting for it reduces odometry drift by a factor of 4.4 (Section III-E).
3) A pre-flight predictor of deployability. The fraction of frames that match the basemap at a known position separates working from failing flights with $\rho = -0.764$ against log error and a threshold near 50%. Because it requires no flight, it turns "will this work here?" into a measurement (Section V-B). The same predictor transfers to a thermal sensor at night, where a structure score read off the basemap alone separates localisable frames from the rest at $\mathrm{AUC} = 0.852$ (Section V-C).
4) A negative result on matcher choice, with the mechanism identified: a stronger dense matcher degrades localization because it lacks the ability to refuse (Section VI).

II. Related Work

Cross-view geo-localization. The dominant formulation matches a single UAV image against a database of satellite tiles. Recent entries include transformer and state-space architectures [1], [2], robustness to off-nadir viewing [3], degraded imagery [4], and metric scale recovery via semantic anchors [5]. Surveying twenty such papers from 2024–2026 we found none that fuse the per-frame result with a motion model over time.

Sequential localization against overhead imagery. Particle filtering against aerial maps is established for *ground* platforms: BEV-Patch-PF combines learned bird's-eye matching with a particle filter for off-road vehicles [6], and earlier work localises street-level imagery city-wide to roughly 20 m [7]. For aircraft the closest work predates modern learned matching: Shan et al. [8] filter optical flow against map correlation. Our contribution sits at the intersection: a modern detector-free matcher inside a sequential filter, on an aircraft, evaluated with metric error over multi-kilometre trajectories.

Image matching. LoFTR [9] performs detector-free coarse-to-fine matching with transformers. RoMa [10] predicts a dense warp with a foundation-model encoder and leads current matching benchmarks. We use both and report in Section VI why benchmark ranking does not transfer to this task.

III. Method

A. Frame preparation

Each UAV frame is rotated to north-up using the platform heading and resampled to the basemap's ground scale, giving a metric, north-aligned crop covering 300×300 m. Heading comes from the inertial unit and altitude from the barometric altimeter; neither depends on GNSS, matching the pose-prior protocol used by contemporary benchmarks. The basemap is stored in EPSG:4326, whose pixels are square in degrees but not in metres — at our latitude 0.2526 m/px east-west against 0.2974 m/px north-south, an 17.8% anisotropy — so satellite crops are resampled to be metrically square before matching.

B. Absolute fixes

A grid of overlapping 300 m basemap tiles is indexed with frozen DINOv2 descriptors [11]. For a query frame, the nearest tiles by cosine similarity give candidate regions; each candidate is verified by LoFTR matching against a 400 m crop, followed by MAGSAC homography estimation. The image centre transported through the homography gives a ground point.

C. From image centre to aircraft position

The matched ground point is where the optical axis meets the ground, not where the aircraft is. With altitude h and attitude angle θ the two differ by $h \tan\theta$; at 466 m a $2°$ tilt displaces the estimate by 16 m. Regressing localization error in the body frame against attitude recovers this relation with coefficients $+0.981$ (along-track) and -0.972 (cross-track), confirming the model. Calibrating the constant mounting offset on a flight's first 20% and evaluating on the remaining 80% reduces median error from 16.82 m to 6.12 m.

Two safeguards proved necessary across flights. First, the recorded height is not always true height above ground; on one flight (recorded 2313 m) applying the correction *increased* error from 27.8 m to 131.2 m, so an effective altitude is estimated from the data. Second, calibration frames are split in half, the correction is fitted on one half and verified on the other, and a correction that does not reduce error is not applied. This declined to fire on two of nine flights.

D. Sequential fusion

State is planar position. The motion model applies the odometry increment with process noise matched to its measured step accuracy. The measurement model treats candidate positions as a multi-modal likelihood with a flat outlier floor. A Kalman filter is inappropriate here: map matching is usually accurate to metres but occasionally confident and completely wrong, giving a heavy-tailed, multi-modal error distribution that a single Gaussian mode cannot represent.

Two mechanisms proved essential in practice. *Measurement-driven injection*: once the particle cloud contracts to a few metres, no particle lies near a fix 100 m away, the likelihood is uniformly zero and the filter cannot move even when the measurement is correct — we observed a filter locked 105 m from truth for twenty frames while receiving 4–10 m measurements. Replacing a fraction of particles with draws from the measurement distribution each step resolves this. *Dead reckoning on the heading*: when odometry fails, propagating the last motion *vector* fails at turns; propagating speed along the inertially-known heading does not.

Because the predicted position is known, verification normally requires a single match attempt rather than a search over the map: 1.52 calls per frame against 3.80 for the single-frame baseline.

E. Self-calibration from the map

The homography aligning a north-up frame to a north-up tile carries a residual rotation. If heading were exact this residual would be zero; it is therefore a direct measurement of heading error. Measured across a survey flight it was $-1.93°$ ($\sigma = 0.86$) on the northwest legs and $-7.08°$ ($\sigma = 1.73$) on the southeast legs. Heading error varying with heading is the signature of magnetometer hard-iron distortion. The GNSS-derived odometry direction bias on the same legs was $+1.65°$ and $+7.22°$: same magnitudes, opposite sign, as the geometry predicts. Feeding a running estimate of this residual back into the odometry reduced drift from 3.795% to 0.865% of distance travelled, that is 2803 m to 639 m over 74 km, using no positioning information of any kind. The scale component of the same homography similarly corrects the assumed ground sampling distance as terrain elevation changes.

IV. Experimental Setup

We use UAV-VisLoc [12], which pairs real survey flights with georeferenced basemaps and post-processed GNSS. We

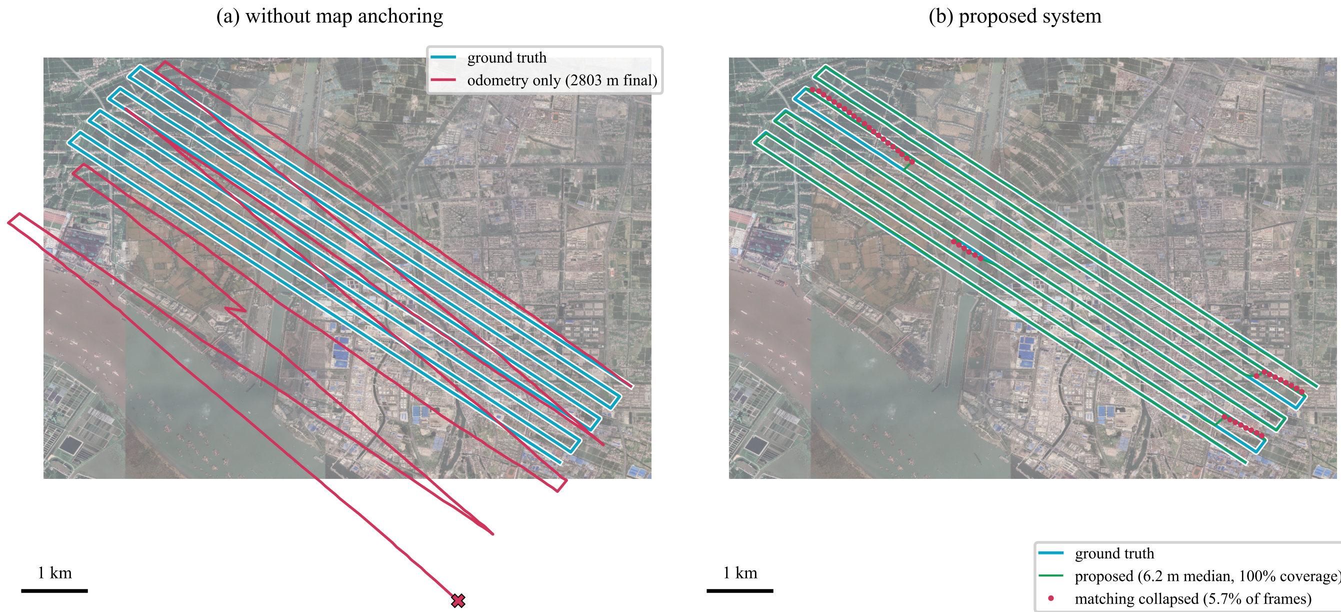

Fig. 1. Flight 03 (768 frames, 74 km, 466 m AGL) over its 8.8×7.3 km orthophoto. Both panels share one frame of reference. (a) Visual odometry alone accumulates heading and scale error until it leaves the basemap entirely, ending 2803 m from truth (×). (b) The same odometry anchored by map matches stays on the true track for every frame; the marked frames are the four segments where matching collapses and the filter coasts on the motion model alone until it re-anchors.

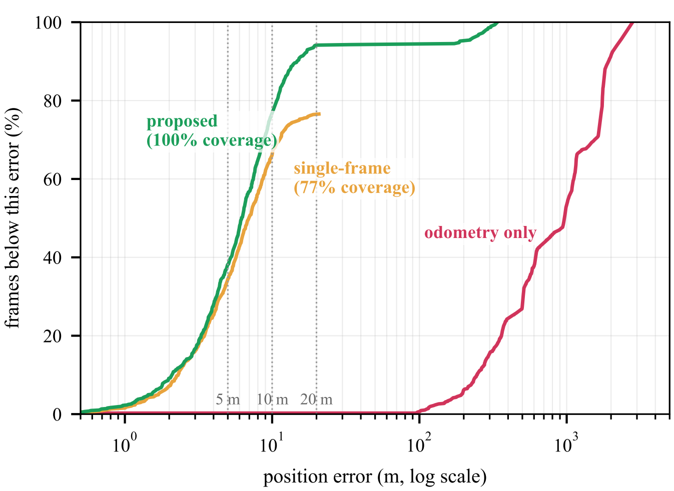

Fig. 2. Error distribution on flight 03. The denominator is every frame of the flight, not every frame a method chose to answer, so a curve that stops short is a method that declined: single-frame matching ends at 77% because it produces nothing on the rest. Fusion converts that shortfall into coverage without giving up the accuracy of the frames matching does solve.

verified the reference quality independently: on straight survey legs the cross-track scatter is 1.50 m and the step-length scatter 0.60 m, so the reference is clean relative to the errors we report.

Of eleven flights we evaluate ten. The excluded one has no attitude or heading columns in its metadata, which the method requires; our loader rejects it explicitly rather than producing unsupported numbers. One flight ships its basemap as a 2×2 grid of GeoTIFFs totalling 4.5 GB rather than a single file; we present these to the pipeline through a GDAL virtual raster so that no copy is made, which is also how tiled basemaps would arrive in deployment. Per-flight ground sampling distance and camera boresight are estimated from each flight's own first 20% and all reported figures come from the remaining 80%.

We report coverage (fraction of frames for which a position is produced), median and 90th-percentile error, and matching calls per frame. All experiments run on a laptop NVIDIA RTX 3050 Ti with 4 GB of VRAM; peak usage is 1.4 GB with LoFTR.

V. Results

Table I gives per-flight results. Seven flights localise with median error between 8.4 m and 24.8 m and coverage at or above 99.7%. Three fail.

A. Component comparison

Fig. 1 shows the two failure regimes on the map itself and Fig. 2 the resulting error distributions. On flight 03, comparing the three approaches directly: visual odometry alone covers every frame but drifts to 2803 m; single-frame map matching achieves 5.46 m median but only on the 76.6% of frames it can solve, at 3.80 matching calls per frame; the fused system covers 100% of frames at 6.20 m median with 1.52 calls per frame. The single-frame baseline produces no gross outliers beyond 100 m — when retrieval proposes the wrong region, verification rejects it and the system reports nothing rather than reporting a wrong position.

B. What separates working flights from failing ones

Performance varies by a factor of 78 across flights, which raises the question of whether the system fails on some flights

TABLE I

PER-FLIGHT RESULTS, FULLY AUTOMATIC. THE SEVEN FLIGHTS ABOVE THE RULE HAVE A MATCH RATE OF AT LEAST 50%; THE THREE BELOW DO NOT.

Flight	Frames	Altitude (m)	Distance (km)	Match rate	Coverage	Median error (m)	90th pct. (m)
03	768	466	74	93%	100.0%	**8.35**	20.03
09	766	546	77	70%	100.0%	**14.94**	59.33
06	344	834	24	76%	99.7%	**15.06**	365.32
04	738	544	83	90%	100.0%	**15.44**	54.64
05	473	2313	30	50%	99.8%	**16.69**	182.51
01	817	406	66	77%	100.0%	**22.51**	114.60
11	590	2572	84	90%	99.8%	**24.79**	424.41
02	1071	406	86	37%	99.7%	53.45	343.39
10	144	773	9	13%	84.7%	126.88	324.98
08	1033	551	103	33%	79.7%	648.49	3785.02

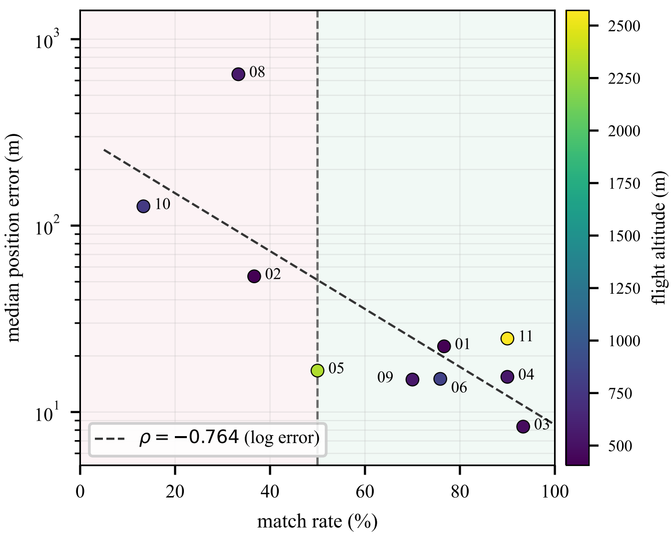

Fig. 3. Match rate against final accuracy across the ten flights. Match rate is measured with the true position given, so it describes the data rather than the algorithm, and it can be evaluated on a planned route before flying. Colour encodes altitude, which does not separate the groups: the 2572 m flight (11) succeeds and the 551 m flight (08) fails.

or whether some flights are unmatchedable. We measure the latter directly: with the true position given, what fraction of frames match the basemap at all? This is a property of the data, since no search is involved.

The answer separates the two groups (Fig. 3). Flights with match rate $\geq 50\%$ give median errors of 8.35–24.79 m (median 15.44 m) at coverage $\geq$99.7%; those below give 53–648 m at 80–85% coverage. The correlation between match rate and log error is -0.764. Altitude is not the discriminator: the 2572 m flight works and the 551 m flight fails.

This is practically useful because match rate needs no flight. Given a planned route and a basemap, it can be measured in advance, converting deployability from a question into a measurement.

C. Does the predictor transfer to another sensor?

The claim behind the match rate is that performance is set by whether camera and basemap depict a recognisably similar world. Nothing in that is specific to visible light, so we tested it on a thermal dataset with a different sensor, terrain and country: Boson-nighttime [13], 26,568 aligned nighttime thermal/satellite pairs over desert, farmland and roads.

The daylight pipeline does not degrade there, it stops. LoFTR on raw thermal returns zero inliers on 100 frames out of 100 — not a worse position, no position. Discarding appearance and keeping structure, by applying CLAHE and a difference-of-Gaussians band-pass identically to both sides, recovers a correct fix on 8.6% of 1000 frames with RoMa.

Which 8.6%? We score each satellite tile for structure using the basemap alone: Canny edge density minus the fraction of spectral energy above an eighth of Nyquist, both standardised. The second term matters because fine texture is not structure — sand speckle and scrub raise the edge count without giving a matcher anything to hold. That score separates the frames that localise from the ones that do not at $\mathrm{AUC} = 0.852$, higher than the inlier count measured *after* matching (0.803). Sorted into deciles, the least structured tenth of tiles produces no correct fix at all; the most structured tenth produces 43%. Used as a pre-filter it also pays for itself in compute: discarding the lower half of tiles before matching retains 90% of the correct fixes.

The predictor therefore survives a change of sensor, and at night it takes a stronger form — it needs only the basemap, not a trial match. It is also worth its place as a component. Combining it with a matcher-side check — accept a fix only where two band-passed representations independently agree to within the tolerance, on tiles in the top 30% by structure — yields 95% precision on 6.2% of frames while issuing 0.6 matcher calls per frame, against 92% precision on 2.4% of frames at 1.0 calls for the tuned single-score gate.

We report this as evidence for the predictor rather than as a night system. A trustworthy fix on 6% of frames is still far below what the daylight flights supply, and the measurements come from a single dataset.

D. Ablation

Removing components one at a time (300 frames, fixed seed): the boresight correction is by far the largest single contributor (17.13 m without it against 6.62 m with); removing odometry barely changes the median but destroys the tail (90th percentile 368 m against 16 m), which is precisely what a motion model is for; removing measurement-driven injection costs 1.0 m median and 3.3 m at the 90th percentile.

Two components do not earn their place and we report this rather than claiming otherwise: particle count is nearly irrelevant (7.08 m with 100 particles, 6.80 m with 2000, 6.62 m with 600), and removing the likelihood outlier floor changes nothing measurable, because gating and injection already filter bad fixes upstream.

E. Robustness

Across twenty conditions on 300 frames, the pattern is that the system is indifferent to radiometry and sensitive to texture. Fog at maximum severity costs 0.6 m; occlusion covering a third of the frame costs 0.6 m; resolution loss costs 0.9 m. Motion blur is the failure mode: moderate blur takes median error from 6.62 m to 58.31 m, and heavy blur to 500 m. Extreme darkness prevents initialisation entirely, producing no position at all.

Discarding satellite fixes at random costs little until they become scarce: 6.62 m at no dropout, 7.38 m with half of all fixes discarded, 10.76 m at 75% and 41.55 m at 90%.

Blur can be partially mitigated by equalising the domain: since the basemap tile is sharp and depicts the same ground at the same scale, the sharpness gap between query and tile measures the blur, and blurring the tile to match reduces median error under sustained blur from 44.47 m to 20.00 m and the 90th percentile from 214 m to 91 m. It does not restore clean performance; heavy vibration remains a genuine limit.

VI. What Makes a Matcher Useful Here

Section V-B identifies match quality as the binding constraint, which suggests replacing LoFTR with a stronger matcher. RoMa [10] leads current matching benchmarks. Measured at known true positions it appears to solve the problem outright: match rate on the three failing flights rises from 12–42% to 96–100%, within 2.7 GB of VRAM.

End-to-end it is far worse: 1669 m on flight 01, where LoFTR gives 22 m.

The bake-off asked the wrong question. It showed each matcher only the *correct* tile, whereas a localization system spends most of its effort deciding whether a candidate is the right place at all. Matching each frame against a crop from a random, unrelated part of the map gives:

	Correct place	Wrong place	Ratio
LoFTR	709 inliers	**0**	709×
RoMa	4598 inliers	341	13.5×

LoFTR returns nothing on unrelated imagery; RoMa returns hundreds of correspondences. Its apparent advantage was an artifact of a metric that only measured the easy direction.

Recalibrating the acceptance threshold does not rescue it. Testing raw inlier count, inlier ratio and matcher confidence on correct-versus-wrong pairs, LoFTR admits a threshold with zero errors under two of the three criteria; for RoMa no threshold exists under any criterion, as some wrong places outscore some correct ones (best accuracy 98.1%).

End-to-end, with thresholds tuned in RoMa's favour, flight 08 makes the cost concrete. LoFTR produces a position on 4.5% of frames, median error 582 m — it is reporting that it does not know. RoMa produces one on 95.5% of frames and is 6250 m wrong on average. In flight the first behaviour triggers a handover to inertial navigation; the second flies the aircraft six kilometres off course without indication.

We therefore keep LoFTR, not because it is the stronger matcher — it is not — but because this task requires a property that matching benchmarks do not measure. A matcher is useful for localization in proportion to its willingness to return nothing.

VII. Limitations

All ten flights come from one dataset and one country, and although altitudes span 406–2572 m and dates 2016–2023, the capture system and basemap class are common to all. Three of the ten fail, and while the cause is measured, it is also a real operational limit. Processing is offline at 866 ms per frame on a laptop GPU; frames arrive every 7 s in this data, so it is comfortably real-time for these flights, but it has not been run on embedded hardware. Attitude and altitude are assumed available from non-GNSS sensors, and Section III-E shows those sensors are themselves imperfect. The planar homography assumption holds for near-nadir frames and degrades with obliquity; building relief displacement is the dominant residual error source and is why the achievable floor is near 6 m rather than near 1 m.

VIII. Conclusion

Treating GNSS-denied visual localization as sequential estimation rather than per-frame retrieval changes which problem is hard. Retrieval accuracy is not the constraint; the constraint is that many frames cannot be matched at all, and a motion model covers exactly those. Across ten real flights the fused system localises 99.7% or more of frames at 8–25 m median wherever the basemap and the camera depict a recognisably similar world, and the fraction of frames satisfying that condition can be measured before flight.

The same match geometry that yields position also yields the platform's own heading error, allowing a magnetometer to be calibrated against the map with no positioning information. And the matcher study suggests a criterion the matching literature does not currently report: for localization, selectivity dominates sensitivity. A matcher that always answers is worse than one that sometimes declines.

Code, data preparation and every measurement reported here are available at https://github.com/YusufGUNEL/GeoAnchor.

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